

Q-1: You will run the following instruction set in a single cycle datapath:

Add X2, X3, X5

Add X4, X2, X7

Sub X5, X2, X3

Ld X2, 12(X8)

Instruction fetch takes 20s

Register read takes 20s

ALU operation takes 40s

Data memory operations take 20s

Register write takes 20s

a) How many clock cycles are necessary to run the above instruction set?

b) What is the total time necessary to run the instruction set?

Q-2: You will run the following instruction set in a pipelined datapath:

Add X2, X3, X5

Add X4, X2, X7

Sub X5, X2, X3

Ld X2, 12(X8)

Instruction fetch takes 20s

Register read takes 20s

ALU operation takes 40s

Data memory operations take 20s

Register write takes 20s

- a) How many clock cycles are necessary to run the above instruction set?
- b) Find if there are any data hazards in the instruction set or not; If yes, solve the data hazard using stalling only.
- c) Find if there are any data hazards in the instruction set or not; If yes, solve the data hazard using stalling+forwarding only.
- d) What is the total time necessary to run the instruction set?

Q-3: You will run the following instruction set in a pipelined datapath:

Add X2, X3, X5

Add X4, X2, X7

Sub X5, X2, X3

Ld X2, 12(X4)

Instruction fetch takes 20s

Register read takes 20s

ALU operation takes 40s

Data memory operations take 20s

Register write takes 20s

- a) How many clock cycles are necessary to run the above instruction set?
- b) Find if there are any data hazards in the instruction set or not; If yes, solve the data hazard using stalling only.
- c) Find if there are any data hazards in the instruction set or not; If yes, solve the data hazard using stalling+forwarding only.
- d) What is the total time necessary to run the instruction set?

Q-4: You will run the following instruction set in a pipelined datapath:

Add X12, X5, X11
Sd X5, 16(X12)
Addi X6, X5, X5
Ld X6, 16(X15)
Sub X8, X5, X6

Instruction fetch takes 20s

Register read takes 20s

ALU operation takes 40s

Data memory operations take 20s

Register write takes 20s

- a) How many clock cycles are necessary to run the above instruction set?
- b) Find if there are any data hazards in the instruction set or not; If yes, solve the data hazard using stalling only.
- c) Find if there are any data hazards in the instruction set or not; If yes, solve the data hazard using stalling+forwarding only.
- d) What is the total time necessary to run the instruction set?